

AUTO DATA STORYTELLER REPORT

Executive Summary

****Executive Summary**** Our predictive analytics initiative has successfully developed a highly effective model with strong classification performance, accurately identifying outcomes in over 83% of cases. This robust predictive capability provides a reliable foundation for understanding key influential factors and guiding strategic business decisions, giving us high confidence in the insights derived. The primary drivers influencing these predictions have been identified as an individual's gender, the price paid for the service or product (fare), and age. Gender, in particular, stands out as the most significant determinant, followed closely by the fare and age. These factors consistently exert the strongest influence on the predicted outcomes. Given the model's strong performance and the clear identification of these major drivers, we recommend close and continuous monitoring of gender, fare, and age trends. Changes or shifts within these critical variables are highly likely to have the most substantial impact on future outcomes, enabling proactive adjustments and more effective resource allocation. Leveraging these insights can significantly enhance strategic planning and operational efficiency.

Dataset Overview

Problem Type: classification

Best Model: Random Forest

Data Quality

- Dataset contains 891 rows & 12 columns
- 866 missing values were identified.
- Column Cabin has the highest missing values (687 values)
- No strong relationships were identified.

Model Findings

CLASSIFICATION MODEL PERFORMANCE

- Random Forest was selected as the best performing model based on F1.
- Random Forest achieved the highest F1 score of 0.83.
- The model correctly classified approximately 83.2% of records.
- The model demonstrates strong classification performance.
- Random Forest outperformed Logistic Regression, Decision Tree based on F1.

Key Drivers

MODEL DRIVERS

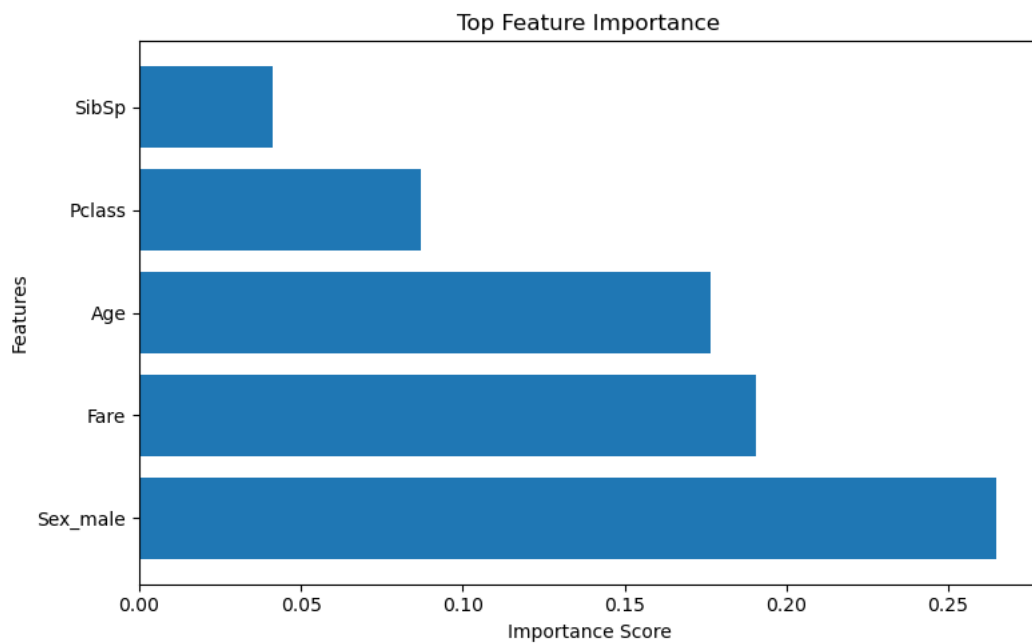
- Sex_male was the most influential feature.
- Fare was among the strongest predictors.
- Age contributed significantly to model decisions.

Recommendations

RECOMMENDATIONS

- Sex_male appears to be the strongest driver of predictions and should be monitored closely.
- The model demonstrates strong predictive performance, making the insights reasonably reliable.
- The most influential features should be tracked regularly, as changes in these variables are likely to have the largest effect on outcomes.

Feature Importance



Missing Values

